Predicting Pedestrian Crossing Intention with Behavioral and Textual Feature Fusion

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Outline

- Introduction
- Method
- Results
- Conclusion

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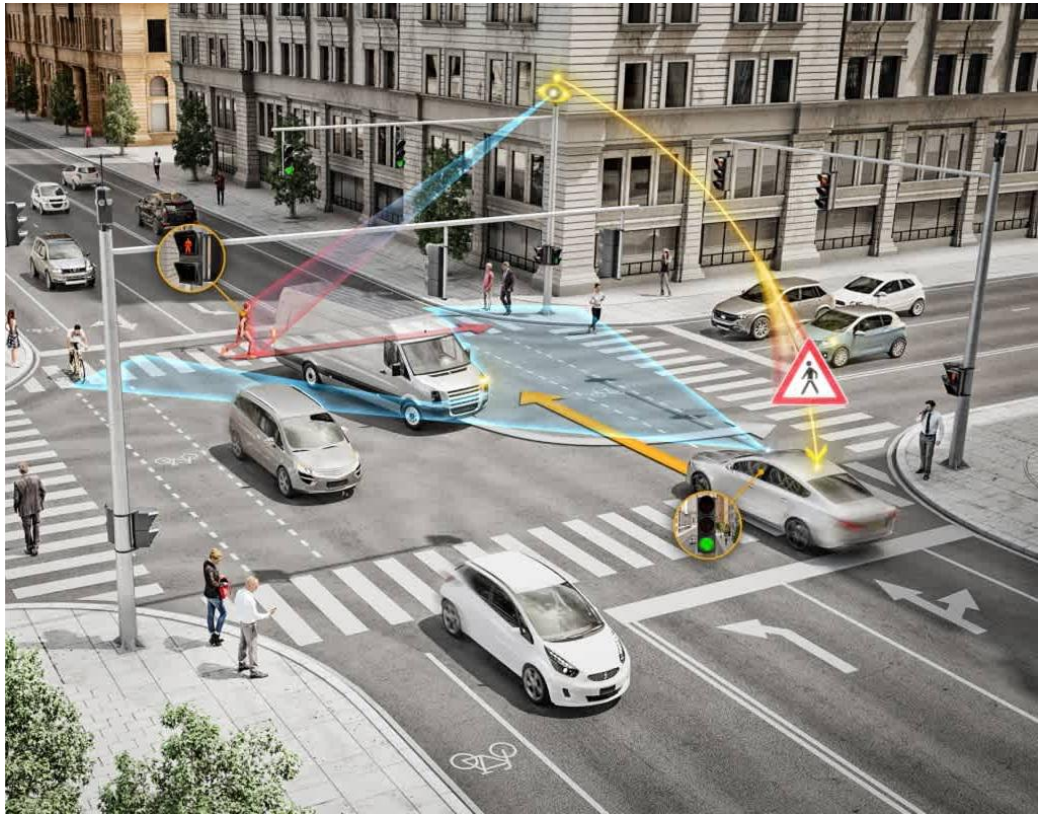
Introduction

Predicting Crossing Intention

- Variable pedestrian behavior
- Multi-modal data
  - visual cues
  - dynamic info
  - environmental context

Importance

- Enhance road safety
- Support autonomous driving



source: https://www.smartcitiesworld.net/news/smart-intersection-aims-to-increase-safety-2422

Introduction

Dataset: JAAD [1]

- 346 short video clips
- pedestrian behaviors
- driver behaviors
- crossing or not

Method

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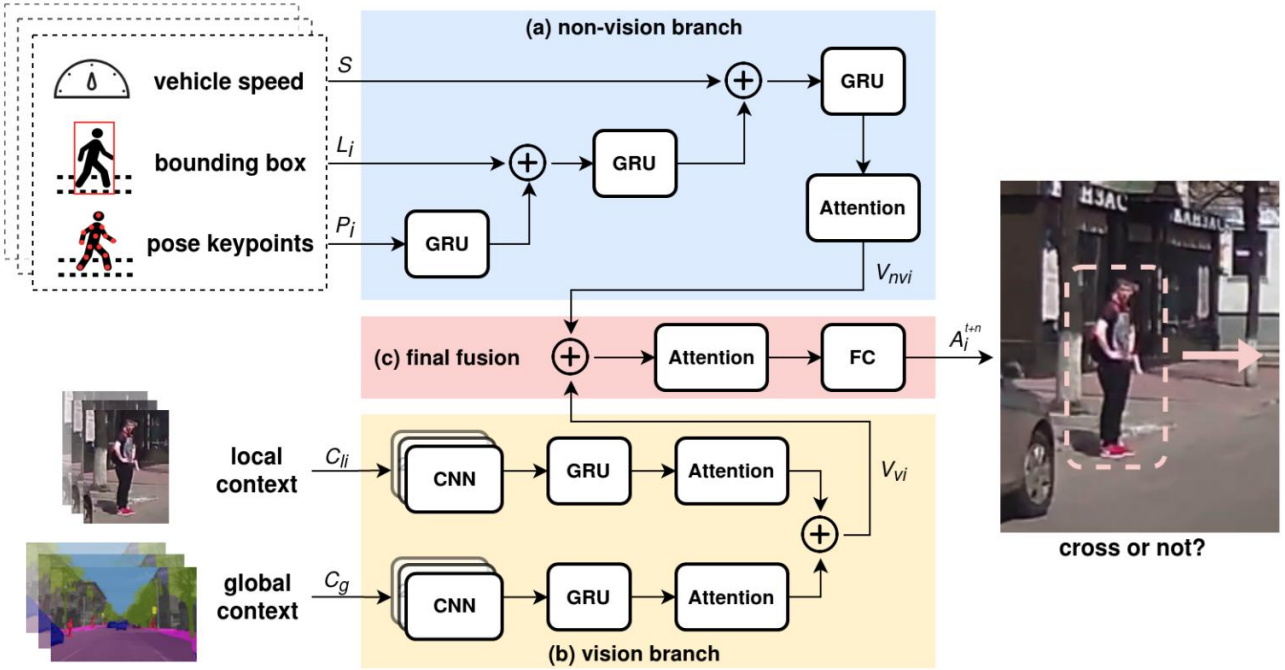
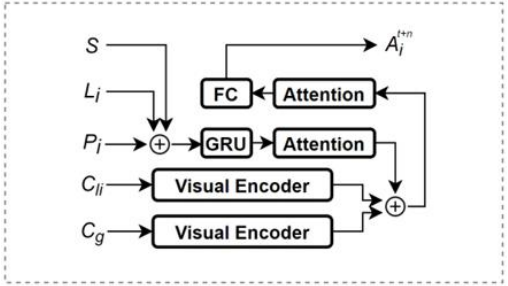
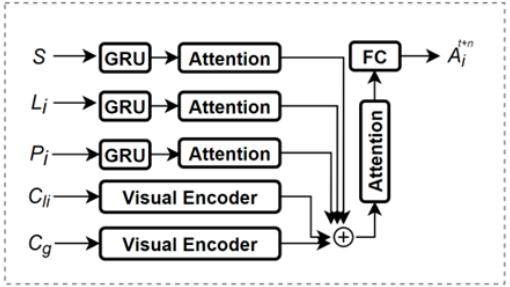
Method





Early Fusion

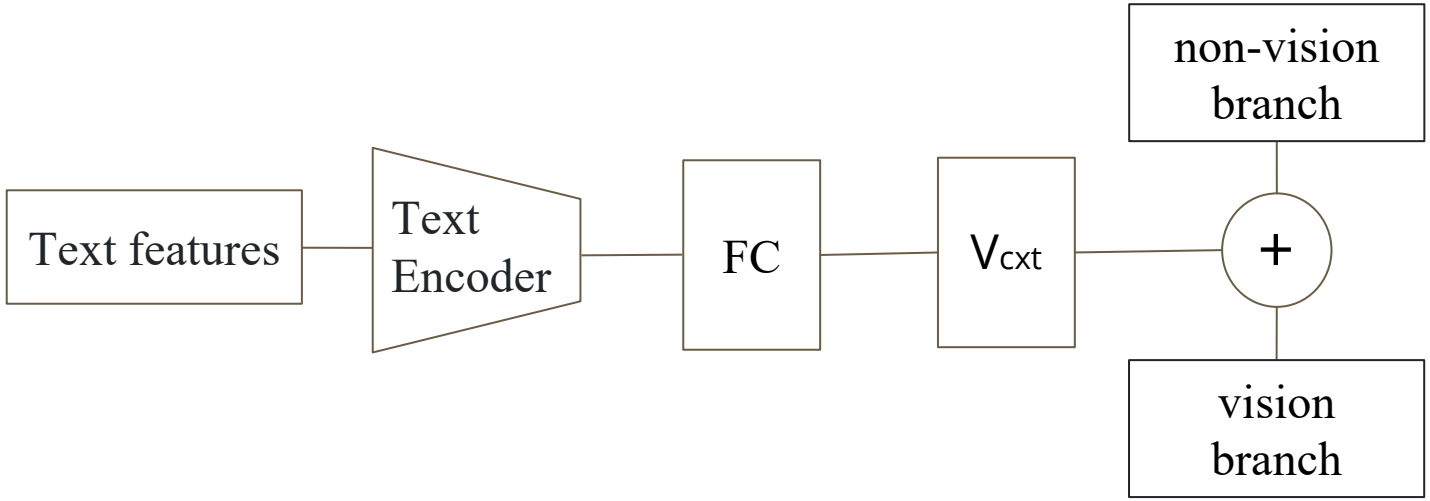


Later Fusion

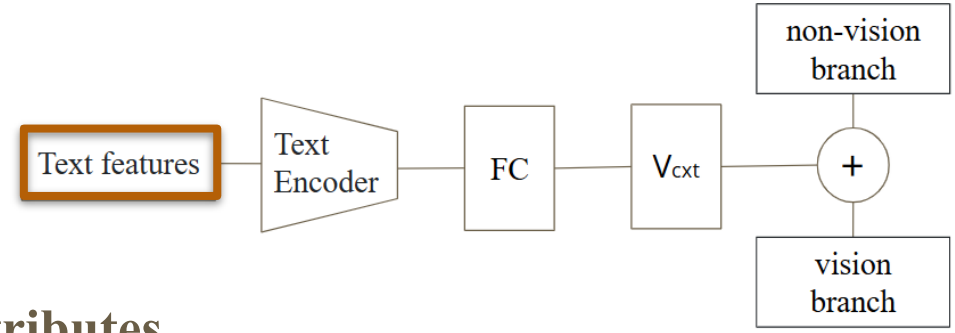
D. Yang et al., "Predicting Pedestrian Crossing Intention With Feature Fusion and Spatio-Temporal Attention" [2]

Method

- Only few features are used in the original method
- More features of the JAAD dataset can be utilized
- Use a text encoder to encode the text features



Method



Behavior

- action
- look

Appearance

- talking on phone

Attributes

- age
- intersection
- number of lanes
- group size
- designated
- signalized
- motion direction

Results

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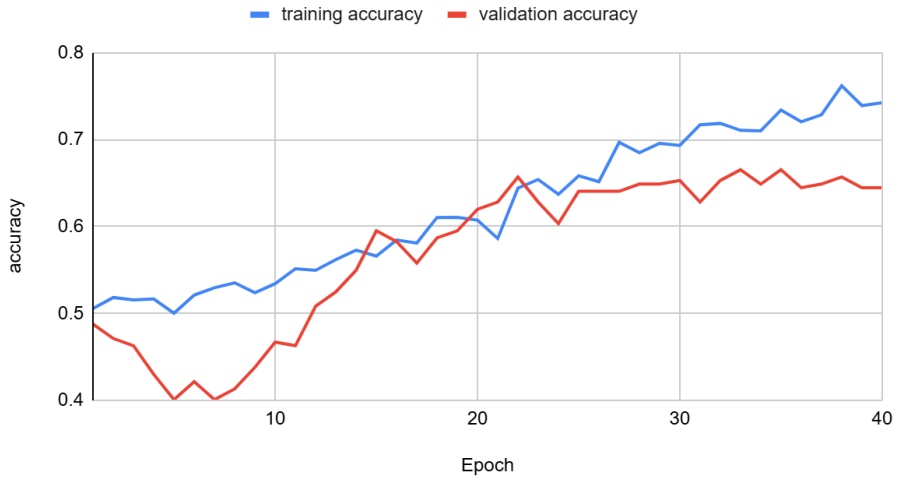
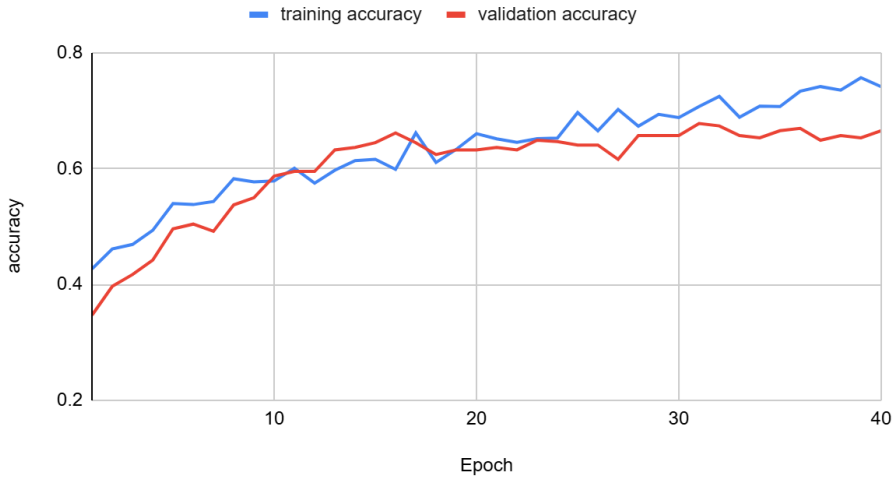
Results

Original laterfusion

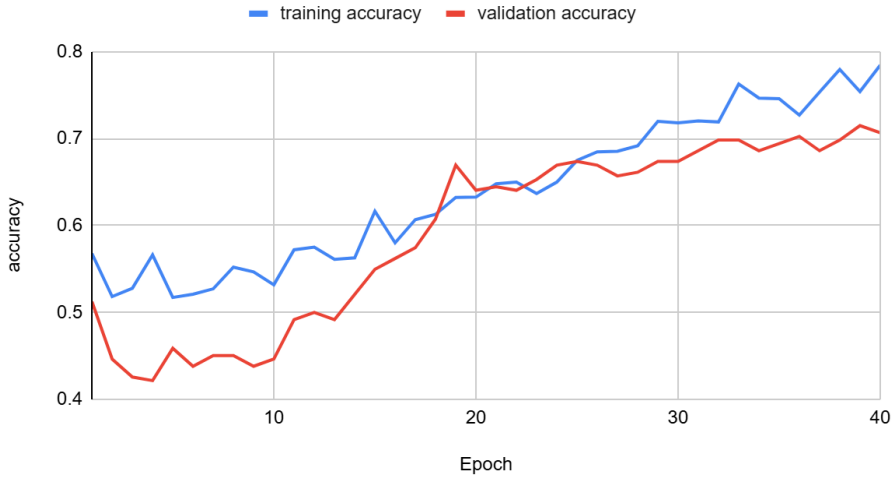


Without text encoder

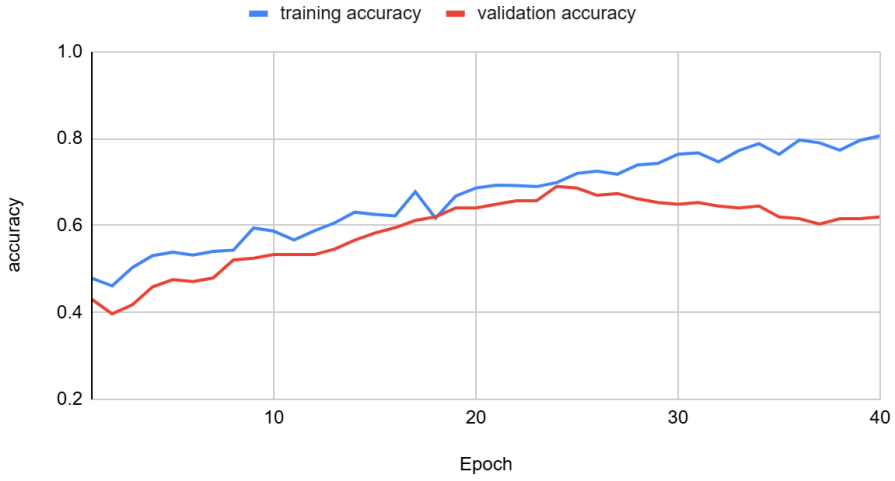


Results

With text encoder (BERT)



With text encoder (DeBERTa)



Results

Method	baseline	No Text Encoder	BERT [3]	DeBERTa [4]
Testing Accuracy	0.66	0.67	**0.71**	0.69

Conclusion

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Conclusion

- Underutilized behavioral and contextual information
- Integration of additional features
- Apply text encoders and attention mechanism
- BERT performs better among text encoders
- Substantial improvements over baseline

Reference

[1] Kotseruba, I., Rasouli, A., & Tsotsos, J. K. (2020). *Joint Attention in Autonomous Driving (JAAD)*. arXiv. https://arxiv.org/abs/1609.04741
[2] Rasouli, A., Kotseruba, I., & Tsotsos, J. K. (2017). *Are they going to cross? A benchmark dataset and baseline for pedestrian crosswalk behavior*. In *Proceedings of the IEEE International Conference on Computer Vision Workshops* (pp. 206–213).
[3] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 4171-4186. Association for Computational Linguistics. https://aclanthology.org/N19-1423/ https://doi.org/10.18653/v1/N19-1423
[4] He, P., Liu, X., Gao, J., & Chen, W. (2021). DeBERTa: Decoding-enhanced BERT with disentangled attention. *arXiv*. https://arxiv.org/abs/2006.03654
[5] Rasouli, A., Kotseruba, I., & Tsotsos, J. K. (2017). *Agreeing to cross: How drivers and pedestrians communicate*. In *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)* (pp. 264–269).
[6] D. Yang, H. Zhang, E. Yurtsever, K. A. Redmill and Ü. Özgüner, "Predicting Pedestrian Crossing Intention With Feature Fusion and Spatio-Temporal Attention," in *IEEE Transactions on Intelligent Vehicles*, vol. 7, no. 2, pp. 221-230, June 2022, doi: 10.1109/TIV.2022.3162719

Thank You